BACKEND IMPLEMENTATION PLAN

Photo Evidence Feature

Project: Foodinesia CCTV-SOP Date: 2026-03-09

Database · Storage · REST API · Socket.IO

🗄 Database Table — reports

TUJUAN

Menyimpan metadata tiap foto tangkapan AI Engine ke database relasional.

SCHEMA

`id`	UUID	Primary key — auto-generated
`filename`	String	Nama file lokal atau path Cloud
`timestamp`	DateTime	Waktu deteksi terjadi
`location`	String	Area / nama kamera sumber
`type`	Enum	'pelanggaran' | 'valid'
`jenis`	String	Deskripsi singkat hasil deteksi
`confidence_score`	Float	Akurasi AI (0.0 – 1.0)

💾 File Storage

OPSI PENYIMPANAN FOTO FISIK

Local — Simpel

Simpan ke folder server: `public/uploads/reports/`

Sajikan via `express.static()`

Cloud — Scalable

AWS S3 / Cloudflare R2 / Minio

File harus punya **public URL** agar bisa dirender oleh frontend.

🔌 REST API Endpoints

ENDPOINT YANG DIBUTUHKAN FRONTEND

POST	`/api/reports`	Terima gambar (Multer) + metadata dari AI Engine, simpan ke DB & storage.
GET	`/api/reports`	Daftar laporan — filter type, pagination. Dipakai Reports.jsx.
GET	`/api/reports/:filename`	Serve foto asli atau redirect ke URL Cloud.
GET	`/api/reports/export/csv`	Generate & download file CSV rekap laporan.

⚠️ **Catatan Keamanan**

Endpoint `POST /api/reports` hanya boleh dipanggil oleh AI Engine dengan header `X-Internal-Key`.

⚡ Real-Time Socket.IO

ALUR BROADCAST DETEKSI BARU

→ AI Engine memanggil `POST /api/reports`
→ Backend simpan data, lalu **broadcast** Socket.IO ke semua client
→ Frontend update Activity Feed & popup foto real-time

PAYLOAD — DETECTION_EVENT

```
{  id, name, status, track_id, timestamp,
   photo_path ← wajib ada
}
```

ALUR DATA END-TO-END

AI Engine → Backend API → DB + Storage → Socket.IO → Dashboard